Global scientific panel on chronic respiratory diseases

Background paper

GLOBAL ALLIANCE AGAINST CHRONIC RESPIRATORY DISEASES

World Health Organization

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Approach to development

This document builds on existing global commitments and strategies relating to noncommunicable diseases (NCDs), highlighting some of the critical issues for chronic respiratory diseases. The background paper presents scientific evidence from published literature and examples from specific projects and countries to support key messages.

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Abbreviations

COPD	Chronic obstructive pulmonary disease
COVID-19	Coronavirus disease 2019
CRD	Chronic respiratory disease
GARD	Global Alliance against Chronic Respiratory Diseases
LMICs	Low- and middle-income countries
NBPHS	National Basic Public Health Service
NCD	Noncommunicable diseases
NGOs	Nongovernmental organizations
PM2.5	Particulate matter 2.5 microns in diameter or less
SABA	Short-acting beta agonists
SDGs	Sustainable Development Goals
UNHLM	United Nations high-level meeting
WHO	World Health Organization
WHO PEN	WHO package of essential NCD disease interventions for primary health care
WHO STEPS	WHO STEPwise approach to NCD surveillance

Context

The 2025 United Nations High-Level Meeting (UNHLM) on the prevention and control of noncommunicable diseases (NCDs) and the promotion of mental health and wellbeing will bring together world leaders to negotiate new commitments to tackle this diverse group of diseases, which threaten life and livelihoods, across all ages, all sectors of society, and all countries.

Chronic respiratory diseases (CRDs) are central to the global NCD agenda and must be part of any NCD response. Worldwide, approximately half a billion people live with CRDs. Asthma, the inflammation and narrowing of lung airways, and chronic obstructive pulmonary disease (COPD), the progressive restriction of lung airflow, are the most common CRDs. Together the two diseases account for almost 4 million deaths every year.

Early diagnosis of asthma and COPD, and access to effective treatment, prevent unnecessary suffering by alleviating daily symptoms, and reduce acute attacks, emergency health care use, and hospital admissions. In addition to healthcare costs, CRDs impose a significant economic impact through lost productivity.

The following five recommendations build on and strengthen commitments from previous UNHLMs on NCDs (in 2011, 2014, 2018) and universal health coverage (in 2019, 2023). Member States are encouraged to:

1. Take action to address chronic respiratory diseases (CRDs)

   Place CRDs at the centre of the national response to NCDs, with commitments to strengthen primary health care and achieve universal health coverage.

2. Foster multisectoral partnerships

   Implement multisectoral strategies to reduce exposure to respiratory risk factors, including tobacco and other environmental factors, across the life course.

3. Strengthen primary care for CRDs to achieve universal health coverage

   Integrate CRDs diagnosis and long-term protection and management into multi-disciplinary primary care systems, including referral for specialist advice and rehabilitation.

4. Ensure access to essential medicines for CRDs

   Ensure inhalers for COPD and asthma are included on national essential medicines lists, are available in primary care, and are affordable to patients.

5. Invest in research and data to drive policy

   Strengthen national and global data systems to track asthma and COPD, including access to diagnosis and treatment, so policies reflect the true health and economic impact.

1. Take action to address chronic respiratory diseases (CRDs)

Place CRDs at the centre of the national response to NCDs, with commitments to strengthen primary health care and achieve universal health coverage.

1.1 CRDs are relatively neglected in the global health and research agenda

Despite the huge burden of deaths and the negative impact on daily life, asthma and COPD are largely neglected within the global health agenda. The general population has a lack of awareness: people do not understand that CRDs are a significant health problem, rating them inaccurately as less harmful and less common than cancer, cardiovascular disease, and diabetes *(1)*. Policymakers also have a low awareness of the adverse effects of CRDs: a 2022 Ipsos poll found that one-third of policy makers did not know that COPD was a lung disease, and only 5% considered it a priority despite being the third leading cause of deaths from NCDs, behind hypertension and stroke *(2)*.

The lack of political interest and allocated resources means that every day, adults and children are suffering needlessly due to lack of diagnosis, and lack of appropriate counselling and treatment.

1.2 Member States have a strong mandate to address CRDs, as part of the NCD agenda

CRDs are central to the global NCD agenda, included as one of the "5x5" major diseases, which share five risk factors of tobacco use, unhealthy diet, physical inactivity, harmful use of alcohol, and air pollution, in the WHO Global Action Plan for the Prevention and Control of Noncommunicable Diseases (2013-2030) *(3)*. Effective prevention, diagnosis, treatment, and monitoring of asthma and COPD are essential to reduce the global burden of deaths and suffering, particularly in low- and middle-income countries (LMICs), and to achieve Sustainable Development Goal (SDG) target 3.4 of a one-third mortality reduction from NCDs *(4)*. Through United Nations and World Health Assembly resolutions, Member States have committed to several global NCD targets, which include CRDs:

WHO Global action plan for the prevention and control of noncommunicable diseases (2013–2030), voluntary global targets (3):

- a 25% relative reduction in risk of premature mortality from cardiovascular diseases, cancer, diabetes, or CRDs;
- an 80% availability of the affordable basic technologies and essential medicines, including generics, required to treat major noncommunicable diseases in both public and private facilities; and
- a 30% relative reduction in prevalence of current tobacco use in persons aged 15+ years.

United Nations 2030 agenda for sustainable development (5).

- Target 3.4: By 2030, reduce by one-third premature mortality from NCDs, through prevention, treatment, and the promotion of mental health and well-being.
- Target 3.8: Achieve universal health coverage, including financial risk protection, access to quality essential health-care services, and access to safe, effective, quality, and affordable essential medicines and vaccines for all.

For CRDs, the basic technologies required for diagnosis are peak flow meters and spirometers. Peak flow meters are included as an essential technology in the WHO package of essential NCD interventions for primary health care (WHO PEN) and should be available in all primary care facilities *(6)*.

Spirometry, a physiological test of the rate of air inhalation and exhalation, is essential for the definitive diagnosis of COPD. Despite being a more complex technology than peak flow meters, it is well-standardised, non-invasive, and has the potential to be affordable even in low-resource settings *(7-9)*. Studies show that the broader use of spirometry, including with low-cost devices, could substantially improve diagnostic accuracy and treatment optimisation in LMICs *(10)*.

Essential medicines for CRDs include bronchodilator and steroid inhalers which are included as core medicines in WHO PEN and on the WHO Model List of Essential Medicines *(6, 11)*. The acute and long-term management of CRDs with these inhaled treatments are highlighted as cost-effective interventions in the "best buys" for the prevention and control of NCDs *(12)*. Treating nicotine dependence with essential medicines will benefit those living with CRD, and also reduce new cases of CRDs, cardiovascular diseases, and cancers *(13, 14)*. In addition to peak flow meters and spirometers, pulse oximeters, which measure the blood oxygen level, are also recognized as essential diagnostic tools for CRDs. The WHO facility-based monitoring guidance includes pulse oximetry as a core indicator for assessing respiratory function and monitoring disease severity, particularly in primary care and emergency settings *(15)*.

1.3 People living with asthma and COPD in less-resourced settings are disproportionately affected

Overall, 90% of premature CRD-related deaths (in people aged below 70 years) occur in LMICs. Lack of diagnostic tools and essential medicines results in asthma and COPD often going undiagnosed and untreated. Health systems in resource-constrained settings are more frequently oriented towards the care of infectious diseases. Greater awareness and resources to treat other lung conditions such as pneumonia and tuberculosis, mean that exacerbations of asthma and COPD may be misdiagnosed and inappropriately treated. Lack of systems to deliver care for chronic conditions result in management of acute exacerbations in emergency rooms without provision of long-term treatments, monitoring, and patient education. This crisis-based approach is neither clinically- or cost-effective.

In order to make progress towards SDG 3.4, reduce by one-third premature mortality from NCDs, it is essential that CRD prevention, diagnosis, and treatment is strengthened, particularly in resource-limited settings.

1.4 NCDs, including CRDs, must be part of strengthened primary health care and universal health coverage

Approximately half a billion people are living with CRDs. The diagnosis and management of CRDs has to be embedded into primary health care, in order to provide quality care that is safe, effective, people-centred, timely, efficient, equitable, and integrated.

The second UNHLM political declaration on universal health coverage emphasized the importance of primary health care as the foundation for achieving universal health coverage, ensuring that everyone has access to essential health services without facing financial hardship *(16)*.

Primary health care is defined as a whole-of-society approach that aims to maximize health through (a) primary care and essential public health functions as the core of integrated health services; (b) multisectoral policy and action; and (c) empowered people and communities. Within this, primary care refers to a key process in the health system that supports first-contact, accessible, continuous, comprehensive, and coordinated patient-focused care *(17)*. Integrating NCDs in primary care is a cost-effective, affordable, and equitable model of care that can reduce death and disability, however limited evidence exists on which models of integrated service design work best and why *(18, 19)*.

1.5 Vulnerable populations are particularly affected by CRDs, with increased exposure to risk factors and inadequate access to diagnosis and treatment

The impact of CRDs is profoundly influenced by social and environmental determinants of health, particularly in conflict-affected and vulnerable populations. Air pollution, tobacco smoke exposure, poor housing conditions, and occupational hazards are key environmental risk factors that disproportionately affect poorer communities, exacerbating the global burden of asthma and COPD *(20)*. This burden falls hardest on those already facing disadvantage, reinforcing cycles of poverty and ill health, and widening existing health and social inequalities *(21)*. Social determinants such as poverty, displacement, food insecurity, and limited access to healthcare services further restrict timely diagnosis and treatment, leading to higher morbidity and mortality rates *(22)*. Studies indicate that forced migration and refugee settings expose individuals to indoor and outdoor air pollution, increasing their vulnerability to CRDs while simultaneously limiting access to essential medicines and diagnostic tools *(23)*.

Populations in conflict zones and humanitarian crises face unique challenges in CRD management, as healthcare infrastructure collapses, medical supply chains are disrupted, and respiratory infections become more prevalent, worsening outcomes for those with pre-existing respiratory conditions *(24)*. Research highlights that war-related environmental hazards, such as exposure to burn pits, chemical agents, and dust storms, have led to an increase in CRD prevalence among displaced populations and military personnel *(25, 26)*. Additionally, children in low-income urban areas, especially in informal settlements, suffer from higher rates of asthma exacerbations due to crowded living conditions, mold exposure, and inadequate healthcare access *(27)*.

2. Foster multisectoral partnerships

Implement multisectoral strategies to reduce exposure to respiratory risk factors, including tobacco and other environmental factors, across the life course.

2.1 Life-course approach and cross-sectoral stakeholder engagement and action

A life-course approach to CRD prevention and care recognizes that risks and vulnerabilities begin early in life and accumulate across the lifespan, requiring targeted interventions at every stage, from childhood to older adulthood *(28)*. Lung health is shaped by early-life exposures, including air pollution, second-hand smoke, and respiratory infections, making early prevention strategies critical *(29)*. Engaging stakeholders across age groups, particularly children, adolescents, and working-age adults, taking account of different levels of literacy and health literacy, is essential to designing effective policies and services. Children with asthma, for example, benefit from school-based education and air quality interventions, while workplace initiatives targeting occupational exposure to dust and chemicals are vital for adults in high-risk jobs *(30)*. Partnerships with civil society organizations, patient advocacy groups, and individuals with lived experience of CRDs ensure that strategies are grounded in real-world needs and contexts, promoting greater equity and social accountability *(31)*. Models of co-created policy development, such as participatory approaches in asthma care planning and national NCD strategies, have demonstrated improved community buy-in and more responsive service delivery *(32)*. The 2024 World Health Assembly resolution (WHA77.2) on social participation for universal health coverage emphasizes the critical role of inclusive, equitable, and diverse stakeholder engagement, including civil society, local communities, youth, and persons in vulnerable situations, in shaping health-related decisions at all levels *(33)*. This global commitment reinforces the need for CRD prevention and control strategies to be co-developed with the involvement of communities and people with lived experience, ensuring responsiveness, accountability, and social trust. In many cultural settings, community leaders, including religious figures, hold significant influence and can play a vital role in shaping public perceptions, supporting behaviour change, and fostering trust in CRD prevention and care initiatives.

2.2 Empowering students, teachers, and parents through early health education

Schools and educational institutions represent critical platforms for promoting respiratory health, raising awareness about asthma and other CRDs. Given that asthma is one of the most common chronic conditions in childhood, early health education can empower students, teachers, and parents to recognize symptoms, avoid triggers, and adhere to treatment plans *(34)*. The Global Asthma Network has successfully used school settings to conduct large-scale epidemiological surveys, helping to identify trends in asthma prevalence, symptoms, and environmental exposures across diverse populations *(35)*. Asthma education for school staff can improve adherence to asthma policies and staff preparedness, with significantly more trained schools implementing written asthma policies and demonstrating readiness to manage asthma attacks, including administering salbutamol, a medication that relaxes and opens up airways, via a spacer *(36)*. School-based asthma

education programmes targeting children have also been associated with improved lung function, fewer days of school absence, and reduced emergency department visits *(37)*. These efforts underscore the importance of multisectoral collaboration between health and education systems to promote lung health from an early age.

A promising example of multisectoral collaboration comes from Western Sydney, Australia, where health services, communication agencies, and schools have partnered to implement peer-led programmes targeting smoking cessation and vaping among adolescents in disadvantaged communities. The Peer Education and Resilience Unit initiative empowers students through co-designed health hackathons, encouraging leadership in health promotion and behaviour change strategies *(38)*.

In addition to asthma awareness, schools offer a unique opportunity for tobacco prevention, given that data show that a significant proportion of smokers initiate tobacco and e-cigarette use during school-age years. Nearly 9 out of 10 adults who smoke cigarettes daily first try smoking by age 18, underscoring the importance of prevention efforts during adolescence *(39)*. The WHO Tobacco- and Nicotine-Free School Toolkit provides a structured approach for creating smoke-free school environments *(40)*. Complementing this, the European Union-funded FRESHAIR4Life project is testing peer-coaching interventions using the "If–Then" implementation intention model, which has shown strong evidence in reducing the number of adolescents beginning smoking *(41-43)*. These efforts underscore the importance of multisectoral collaboration between health and education sectors to foster healthy behaviours and promote lung health from an early age.

2.3 Workplaces as settings for CRD prevention and management

Workplaces offer an essential setting for the prevention, early detection, and management of CRDs, particularly in high-risk sectors such as mining, construction, and manufacturing. Programs targeting early detection of occupational lung diseases, such as COPD, pneumoconiosis, and silicosis, have shown value in identifying at-risk workers and preventing disease progression through the conduct of routine spirometry and symptom screening in mining communities and other labour-intensive industries *(44, 45)*. Employers also play a critical role in implementing workplace-based health interventions, including smoking cessation programs, indoor air quality improvements, and the provision of personal protective equipment to reduce exposure to harmful substances like dust, fumes, and chemical agents *(46)*. Evidence shows that workplace-based smoking cessation programs can be particularly effective, benefiting from peer support and employer incentives *(47)*. Occupational health services provide a valuable platform for outreach, education, and linkage to care, especially in LMICs where access to primary care may be limited, as demonstrated by initiatives like "Your Lungs at Work", which raises awareness about workplace respiratory health and prevention strategies *(48)*. Engaging employers in CRD prevention efforts is not only a public health priority but also a matter of social responsibility and long-term economic benefit *(49)*.

2.4 Environmental risk factors like air pollution and climate change contributing to CRDs

Environmental risk factors, particularly air pollution and climate change, are major contributors to the global burden of CRDs. Both indoor and outdoor air pollution have well-established links with

asthma, COPD, and other lung conditions. Key sources include exposure to biomass smoke from cooking and heating, industrial emissions, vehicular traffic, and urban pollution, which disproportionately affect populations in LMICs and marginalized urban communities *(50, 51)*. Nearly half of global COPD deaths are attributable to air pollution, alongside significant proportions of deaths from asthma, lung cancer, heart disease, and stroke *(52)*. Indoor and outdoor air pollution, especially fine particulate matter, 2.5 microns in diameter or less (PM2.5), and ground-level ozone, a gas that can aggravate breathing symptoms and damage lungs, are now recognized as major causes and exacerbators of CRDs. These pollutants also disproportionately affect children, older adults, and marginalized populations, intensifying health inequities *(53)*. Despite a clear mandate from the 2018 UNHLM on NCDs, work needs to be done to better integrate air pollution into the WHO '5x5' NCD framework *(54)*. The 2nd Global Conference on Air Pollution and Health convened by WHO, held in March 2025 in Cartagena, issued a strong call to action for all sectors, including a pledge to reduce the health impacts of air pollution by 50% by 2040 *(55)*. Achieving this target will require global commitment and coordinated action across health, environment, energy, and transport sectors.

The updated WHO Global Air Quality Guidelines provide critical thresholds for pollutants such as PM2.5, nitrogen dioxide, and ozone, and urge governments to integrate air quality management into national health strategies *(56)*. The Low Emission Zone in London, United Kingdom of Great Britain and Northern Ireland, where drivers whose vehicles do not meet established emission standards are charged, has demonstrated significant reductions in vehicle emissions and improvements in urban air quality, particularly in nitrogen dioxide concentrations *(57)*. Building on its success, the city implemented the Ultra Low Emission Zone, which further expanded these gains and now serves as a compelling success case for other densely populated cities grappling with air pollution *(58)*. Several other cities and countries have shown notable progress. The China Center for Disease Control and Prevention is leading the 'Battle for Blue Skies' campaign, which has driven measurable improvements in air quality, particularly in major urban centers *(59)*. In India, the Indo-Gangetic Plain regional air quality investment programme, supported by the World Bank, aims to reduce air pollution across multiple sectors and borders *(60)*. Paris, France, has implemented car use restrictions and expanded green urban policies to lower vehicular emissions and improve health outcomes *(61)*. In Cameroon, youth-led initiatives such as those under the Healthy Lungs for Life campaign are driving community-level change and awareness on air pollution and respiratory health *(62)*. These examples, now linked to reductions in NOx levels, gases that contribute to smog, acid rain, and ground-level ozone, and emergency asthma admissions, highlight the potential of cross-sectoral and policy-driven action to protect respiratory health and reduce CRD burden in both high- and low-income settings

Climate change further compounds respiratory risks through the increasing frequency and intensity of wildfires, sandstorms, and extreme heat events, all of which exacerbate respiratory symptoms and trigger acute attacks in individuals with pre-existing CRDs *(63, 64)*. Notably, the rise in large-scale wildfires has been linked to surges in emergency department visits and hospital admissions for asthma and COPD, even affecting populations living far from the fire origin *(65)*. In addition, exposure to sand and dust storms has increased by 31% over recent decades, with nearly 3.8 billion people globally now annually exposed to dangerous PM10 concentrations, small particles of dust and smoke that have a diameter of 10 microns or less, substantially increasing the burden of lung disease *(66)*. Studies indicate that poor air quality increases the risk of displacement, while forced migration and refugee settings simultaneously expose individuals to indoor and outdoor air pollution, increasing their vulnerability to CRDs and limiting access to essential medicines and diagnostics *(67, 68)*. Addressing these environmental determinants of lung health requires transdisciplinary collaboration and a multisectoral policy response, including embedding health

considerations into national energy and transportation strategies to reduce pollution at the source *(69)*. Transitioning to clean energy is critical not only for climate resilience but also for reducing exposure to harmful air pollutants that contribute to the burden of CRDs. A compelling example is rapid transition to electric vehicles in Nepal, driven by lower import taxes, abundant hydropower, and reduced operating costs, which offers a promising model for LMICs aiming to cut air pollution and improve lung health *(70, 71)*. Expanding access to eco-friendly, clean cookstoves, especially in rural and low-income households, can significantly reduce household air pollution, a major factor in COPD, and childhood asthma. The WHO estimates that transitioning to clean cooking solutions could prevent millions of premature deaths annually, highlighting the need for sustained investment and policy support in clean household energy initiatives *(72)*.

2.5 Tobacco control as a cornerstone of CRD prevention

Tobacco control is a cornerstone of CRD prevention and demands a coordinated multisectoral response beyond the healthcare sector. Effective strategies require robust regulatory, fiscal, and legislative actions, including taxation, advertising bans, standardized packaging, and restrictions on tobacco industry interference, which includes economic power, lobbying and marketing, and media manipulation. The WHO Framework Convention on Tobacco Control, the first global public health treaty, and its MPOWER package provide comprehensive guidance on how countries can reduce tobacco through evidence-based policy measures *(73, 74)*. Notably, legislative interventions such as implementing smoke-free public spaces, raising tobacco excise taxes, and enforcing bans on tobacco marketing have proven among the most cost-effective public health tools to reduce tobacco consumption and second-hand smoke exposure *(75)*. The urgency of greater promoting and enforcing comprehensive smoke-free environments is underscored by the fact that only four countries have implemented all MPOWER measures at best-practice level, while regulatory loopholes continue to be exploited, particularly in LMICs *(76)*. Tobacco use remains one of the leading risk factors for NCDs globally, responsible for approximately one in six NCD-related deaths *(77)*. Strong evidence shows that tobacco taxes are among the most effective tools to reduce smoking rates and prevent NCDs; for instance, in the Philippines, smoking prevalence fell by one-third between 2009 and 2021 following substantial tobacco tax reforms *(78)*. However, global implementation of tobacco taxes remains low, with only 13% of the world's population protected by taxes at recommended best-practice levels *(78)*. These environmental and policy-level actions not only safeguard the general population from harmful exposures but also reduce the environmental footprint of tobacco by lowering demand for land and water used in tobacco cultivation and processing. These environmental and policy actions support SDG 6, to ensure access to water and sanitation, and SDG 15, to conserve life on land *(79)*.

Alarmingly, the use of Electronic Nicotine Delivery Systems (ENDS) is spreading rapidly, particularly among adolescents, with studies linking their uptake to increased risk of nicotine dependence and transition to combustible tobacco use *(80)*. The growing appeal of new tobacco industry products, such as e-cigarettes and heated or heat-not-burn tobacco products which allow the inhalation of nicotine, represents a significant threat to youth with aggressive marketing strategies targeting adolescents. While evidence linking these products to asthma and COPD is still emerging, their rising use highlights the urgent need for stronger preventive efforts, especially within school-based programs. Recent legislative actions in Belgium to strengthen the regulation of e-cigarette sales offer a promising policy model *(81)*. Moreover, continued collaboration between governments and civil society organizations, including nongovernmental organizations (NGOs), typically non-profit entities that work independently of government control, is essential to sustain progress in tobacco control.

In the context of declining funding opportunities for NGO-led prevention projects across parts of the European Union, renewed political and financial support is needed to ensure the continuity and impact of these critical initiatives *(82)*. At the same time, the rising popularity of waterpipes, hookahs, and other non-cigarette tobacco products poses additional risks for CRDs and must be addressed through comprehensive tobacco control policies and public education campaigns *(83-85)*. Strong tobacco control measures, embedded within national development strategies, are vital to curbing the global burden of CRDs such as COPD and lung cancer.

2.6 Urban planning and the role of healthy cities in promoting respiratory health

Urban planning plays a critical role in shaping health-promoting environments that support respiratory health and overall well-being. Action is needed to create more well-designed cities that can reduce exposure to harmful pollutants and facilitate healthy behaviours such as active travel by ensuring access to clean public spaces, expanding green areas, and investing in safe, efficient, and low-emission public transportation systems *(86, 87)*. The integration of health objectives into urban policy, including housing standards, energy systems, and infrastructure development, is essential to address the root causes of CRDs. Poor urban design, inadequate ventilation in housing, and proximity to high-traffic roads are documented to exacerbate exposure to indoor and outdoor pollutants, contributing to the burden of asthma and COPD, particularly in low-income and marginalized communities *(88)*. As emphasized in the Marmot Place framework, building healthy and sustainable cities is central to reducing health inequalities, including those affecting respiratory health, and promoting environmental justice; the eight Marmot principles underscore that social and environmental equity must go hand in hand to improve lung health outcomes at scale *(89)*. Promoting healthy cities requires cross-sectoral collaboration between health, transport, housing, leisure, and environmental agencies to align urban planning with public health priorities. The WHO Healthy Cities framework underscores the need to embed health in all policies and highlights the benefits of inclusive, equitable urban development for respiratory and planetary health alike *(90)*.

2.7 Vital role of the private sector in advancing CRD prevention and care

Engaging the private sector, particularly pharmaceutical and medical device companies, is vital to advancing CRD prevention and care. These industries contribute significantly through the development and distribution of diagnostics, inhaled therapies, updated vaccines, and digital health tools that support disease control and self-management *(91)*. However, high treatment costs and unequal access to essential medicines remain persistent challenges, especially in LMICs *(92, 93)*. The development of public-private partnerships can help overcome these barriers by supporting affordable pricing models, promoting lower cost generic alternatives, and ensuring equitable access through clear regulatory pathways *(94)*. Ethical engagement is also essential to prevent excessive marketing and over-diagnosis, and align commercial practices with public health goals. Strategic collaboration with the private sector can enhance CRD response efforts, foster innovation, and expand access to life-saving interventions.

3. Strengthening primary care for CRDs to achieve universal health coverage

Integrate CRDs diagnosis and long-term protection and management into multi-disciplinary primary care systems including referral for specialist advice and rehabilitation.

3.1 Potential of primary care to deliver quality respiratory care

Primary care has the potential to deliver good quality respiratory care that meets the needs for the majority of people over their lifetime *(95-98)*. It can offer the elements that matter to patients: accessible timely first contact care; continuity through trusted relationships and electronic records; coordination of services for those people with multiple conditions and needs for specialist advice or rehabilitation; comprehensive holistic care, including parity of esteem (referring to the principal of assigning equal value) for both physical and mental health; and people-centredness, empowering people to take an active role in their health *(99)*. However, the public does not yet have timely access to good quality primary respiratory care in many countries across all income levels, and therefore they are deprived of essential treatments and good quality and productive lives *(100)*. The integration of CRD diagnosis, protection and long-term management into multi-disciplinary primary care systems can address this, but it requires strengthening of primary care. Strengthening involves investment in well-trained healthcare teams competent to deliver the basics in asthma and COPD and to know when and how to refer for specialist advice and rehabilitation; investment in technology to support diagnosis and for record-keeping; routine and reliable primary care access and permission to use and prescribe vaccines and medicines; and policies that offer appropriate incentives and rewards *(101)*. Competence rather than profession should drive care delivery, enabling a competence-based skill-mix. Evidence shows that teamwork and task-shifting between professionals can improve efficiency and access but is often implemented badly, taking insufficient account of time, workload, competence, power dynamics or the capacity of managers and supervisors. Investment in clinical teaching is essential to enable peer teaching and learning that uses evidence-based teaching techniques as well as evidence-based content, paying attention to how and what is taught. Content requires the inclusion of teaching of “soft skills” such as empathetic listening of people with respiratory symptoms, in addition to technical skills *(99, 101, 102)*.

3.2 Value of immunization against common respiratory infections for CRD patients

Immunization against common respiratory infections is valuable to protect people diagnosed with CRD and reduce the impact on health systems and societies. People with CRD and other NCDs are more susceptible to severe illness, hospitalization, and death from respiratory infections than people without NCDs. Infection may also worsen their CRD in the short-term and frequently in the long-term *(103-105)*. However, the risk of serious impact can be significantly reduced through immunization, bringing personal and economic benefits; WHO recommends both influenza and

COVID-19 vaccination as Best Buys *(106, 107)*. However, availability and uptake of immunization is variable and requires governments and clinicians to undertake concerted efforts to improve consistent, predictable, and appropriate supply and advocacy of the benefits. During the COVID-19 pandemic primary care, including community pharmacists, demonstrated their high value in increasing immunization uptake *(108)*.

People with COPD are highly likely to have other health problems such as cardio-metabolic disorders, pain, depression, anxiety, cancers, substance misuse, obesity, underweight, low muscle mass, and frailty. These health problems increase with age and each problem increases their risk of poor quality of life, hospital admission, and premature death *(109-111)*. Therefore they benefit from generalist, multidisciplinary services such as primary care and services for older people, which actively identify and manage comorbid conditions and also address risks such as polypharmacy *(112)*. Home and centre-based rehabilitation programmes are effective in improving strength and coping skills that can enable people with CRD and other conditions to breathe better, do more, and live well *(113-117)*. These programs need to be set up in collaboration with the community to ensure they are accessible and culturally relevant, have effective referral programmes from primary care, and integrate other interventions such as falls prevention and medicines management.

3.3 Importance of quality care for achieving universal health coverage

Dr Tedros Adhanom Ghebreyesus, WHO Director-General, has emphasized the importance of quality care, stating, "*Without quality, universal health coverage remains an empty promise. Even with increased access to services, health improvements can remain elusive unless those services are of sufficient quality to be effective (118)*." Therefore, in addition to integrating CRD into primary care services, and equipping and incentivising teams, there is a critical need to establish quality assurance programmes *(119)*. These should include peer-led education, quality standards, and audits to assure the sustainable delivery of good quality primary care *(120-122)*. A core element, which is often missing from primary care services, is agreement about the treatment of asthma and COPD and regular review of personal action plans to save lives and reduce the need for acute care, which is typically more expensive and more distressing for patients *(123, 124)*. While there is global variation in preferences for in-person, live online, and digital services, there is increasing call for good quality information for patients provided by trusted health care providers, and primary care should be required to provide this in a patient-centred way that takes account of general and "breathing" literacy, and also evaluate its impact *(125-128)*.

3.4 Prioritizing valuable actions in primary care, for example addressing tobacco dependence

Primary care time is limited, and therefore it must prioritise the most valuable actions that only primary care can deliver *(129)*. Tobacco smoking is the global leading cause of premature death and disability including CRD, killing almost half its users *(130)*. To improve the effectiveness of primary care, it is essential to include the diagnosis, management, and recording of tobacco dependence in primary care schedules including the prescribing of pharmacotherapy *(131)*. The behavioural change skills needed for cessation counselling, are essential for most CRD care, including shared decision-making for CRD treatments, improving adherence to inhaled medicines, encouraging acceptance of

rehabilitation programmes, and increasing physical activity *(132)*. Tailored educational interventions, particularly for paediatricians, are also needed to enable healthcare professionals to actively promote smoking cessation, as shown in initiatives such as the GARD-Italy demonstration project *(133)*. Therefore, teaching every clinical undergraduate how to ask, advise, and act to address tobacco smoking should be an essential element of all undergraduate teaching programmes.

As part of the global Asthma Right Care initiative, community pharmacist societies in Spain have played a key role in promoting the appropriate use of inhaled medicines. Their efforts have focused on reducing inappropriate over-the-counter supply of short-acting beta agonists (SABAs), which provide quick relief for asthma symptoms, and encouraging the use of inhaled corticosteroids for longer-term asthma management. Key success factors include providing incentives for pharmacists to counsel patients, ensuring access to affordable controller medications such as inhaled corticosteroids, and fostering collaborative relationships with prescribers to support continuity of care *(134)*. The role of primary care should also include educating patients with CRDs on strategies to reduce personal exposure to outdoor air pollution. Evidence-based recommendations, such as those outlined in the 2021 American Thoracic Society Workshop Report, emphasize the importance of tailored interventions, including behavior change, the filtration of indoor air, and real-time air quality awareness *(135)*.

4. Ensure access to essential medicines for CRDs

Ensure inhalers for COPD and asthma are included on national essential medicines lists, are available in primary care, and are affordable to patients.

Essential medicines for CRDs include inhalers to treat asthma and COPD, medicines to help people quit tobacco, and medical oxygen. These are all included in the WHO Model List of Essential Medicines.

4.1 Inhaled medicines for asthma and COPD are a vital component of asthma and COPD treatment, without which patients will suffer increased symptoms, hospitalization, and death.

Inhaled medicines are the cornerstone of treatment for asthma and COPD. Inhaled steroids are proven to reduce airway inflammation and acute attacks *(136-139)*, and reduce the risk of death from asthma *(140)*. Global COPD and asthma guidelines recommend specific inhalers depending on disease severity, including appropriate inhaled bronchodilators, corticosteroids, and combinations of these two, for optimal disease management.

There are wide gaps in access to recommended inhalers, particularly in LMICs, which contributes to the heavy burden of COPD and asthma deaths and disability. The largest study of the availability and affordability of inhalers for COPD and asthma across sixty LMICs *(141, 142)* found that:

- Combination inhalers for asthma and long-acting inhalers for COPD were available and affordable in only 5% of healthcare facilities;
- Inhaled corticosteroids for asthma were available and affordable in one-third of pharmacies and healthcare facilities;
- "Spacers" - plastic tube-shaped devices that make it easier to take asthma medicine (especially for children) are also widely unavailable across LMICs *(143)*; and
- Where inhalers were available, they were often old formulations and/or unaffordable, costing more than a week's wages for a month's supply.

Evidence from Brazil demonstrates the importance and cost-effectiveness of affordable therapy. After free asthma inhalers were introduced, household costs fell from 29% of income to 2% and the hospitalization rate fell from 90 per 100 000 to 60 per 100 000 people *(144)*.

When recommended inhalers are unavailable, patients often rely on less effective and potentially harmful treatments. A study of asthma treatment across 25 countries found widespread use of inappropriate oral medicines, which increased the risk of adverse effects *(145)*.

Action is needed to overcome patients' misperceptions regarding the need for medication, stigma about the use of inhalers, poor adherence to treatment regimens, and incorrect inhaler technique *(146, 147)*.

National-level data on the number of people with COPD and asthma would allow the generation of demand forecasts for inhaled medicines and facilitate the development of mechanisms for calculating total demand. Access to essential medications must also be supported by guidelines on the management of CRDs that ensure patients are prescribed the correct drugs in devices they can use properly. A recent study found that just 22% of LMICs had national COPD guidelines *(148)*.

It is also essential to address the fact that access to inhaled medicines in LMICs is under threat by new European regulations to phase out the most commonly used inhaler which contains fluorinated gas propellants that have the potential to contribute to global warming *(149)*. These inhalers are currently the only feasible option available to most people with asthma in LMICs, especially children. Although ameliorative measures are envisaged by regulators, unless the process is carefully monitored, interruption of supply could result in increased death and disability, especially in LMICs *(150)*. It is vital that supplies of current inhalers remain available until inhaled medications with low global warming potential become available in sufficient quantities and at affordable prices for all patients in LMICs.

Access to essential medication could be improved in a number of ways *(151)*:

- Facilitation of product registration and regulatory harmonization across countries;
- Partnerships with pharmaceutical companies to increase access to inhalers, including by:
    - technology transfer
    - voluntary licensing agreements
    - tiered/differential pricing;
- Regional partnerships for pharmaceutical and health technology manufacturing;
- Price reductions for bulk purchasing/pooled procurement; and
- Promoting cheaper generic alternatives.

4.2 Medicines to treat nicotine dependence can benefit people living with asthma and COPD, and help prevent the development of other NCDs

New medicines to help smokers quit tobacco were added to the WHO Model List of Essential Medicines in 2021. These medicines (bupropion and varenicline) that modulate the effects of nicotine withdrawal are included as effective interventions in the WHO “Best buys” for the prevention and control of NCDs and in recently published WHO clinical treatment guidelines *(152)*. The use of nicotine replacement therapy, bupropion, and varenicline increases the likelihood of quitting, with greatest success seen when these medicines are combined with behavioural interventions *(153)*. Quitting smoking is the most important action that a person with COPD can take to improve their lung health, reducing daily symptoms and COPD attacks *(154)*. It is therefore essential that people with COPD who smoke are supported by healthcare providers, with access to essential nicotine dependence medicines in primary care.

4.3 Ensuring access to medical oxygen for management of acute asthma and COPD attacks

Ensuring access to medical oxygen is a critical component of CRD management, particularly during acute attacks and for patients with advanced COPD. Yet, oxygen remains unavailable or unaffordable in many LMICs, contributing to avoidable morbidity and mortality. The Lancet Global Health Commission on Medical Oxygen highlighted profound gaps in availability, infrastructure, and training, calling for urgent investment to integrate oxygen delivery into primary and secondary care systems, which provide specialist care usually after a primary care referral *(155)*. Strengthening supply chains, ensuring appropriate technologies, and including oxygen in essential health benefit packages are necessary steps to make this life-saving therapy universally accessible *(156)*.

4.4 Urgent need for action to improve access to essential medications

Widespread lack of access to inhalers is undermining achievement of the Global NCD Action Plan target of at least 80% of health facilities with recommended NCD medicines and SDG target 3.8 on universal health coverage, including access to safe, effective, quality, and affordable essential medicines for all. The urgent need for action to improve access to essential medications is emphasized by data in the annex to the Report of the UN Secretary-General on "*Progress on the prevention and control of non-communicable diseases and the promotion of mental health and well-being*" which shows that 4 out of every 10 countries do not have either guidelines or essential medicines lists for the management of COPD and asthma *(157)*.

Improving access to inhaled therapies for asthma and COPD requires not only inclusion on national essential medicines lists, but also the creation of deliberate strategies to ensure affordability and availability in primary care. In many LMICs, the cost of inhalers remains a key barrier to access, particularly where health systems rely heavily on out-of-pocket expenditure of patients for care. National governments and health insurers can play a pivotal role in addressing this challenge by engaging in structured price negotiations with pharmaceutical companies, leveraging pooled procurement mechanisms, and pursuing health system financing reforms that prioritize equitable access *(158)*. Studies have shown that such strategies, when framed around public health goals, can lead to significant reductions in medicine prices and improve availability of essential respiratory therapies in public-sector formularies *(159)*. Establishing transparent pricing frameworks, encouraging generic market competition, and including inhaled corticosteroids and bronchodilators in public insurance schemes can further reduce the economic burden on patients and support long-term disease control.

5. Invest in research and data to drive policy

Strengthen national and global data systems to track asthma and COPD, including access to diagnosis and treatment, so policies reflect the true health and economic impact.

5.1 The importance of research and data collection

Research and data collection are fundamental pillars for the effective prevention, diagnosis, and management of CRDs, in line with objective 5 of the WHO Global Action Plan for the Prevention and Control of NCDs, which urges Member States to promote and support national capacity for high-quality research and development *(3)*. Despite affecting nearly 500 million people worldwide *(160)*, asthma and COPD remain under-researched compared to other NCDs. Notably, COPD ranks as the third leading cause of NCD burden, following heart disease and stroke *(161)*. These findings highlight the urgent need to develop robust surveillance systems that track prevalence of disease and delivery of care, yet significant knowledge gaps, limited disease prevalence data, and suboptimal implementation of cost-effective interventions persist. Strengthening CRD research and surveillance efforts is crucial for driving evidence-based policymaking and promoting equitable access to care globally. Despite the substantial burden of CRDs, accounting for approximately 10% of all NCD-related deaths in 2021 *(160)*, respiratory diseases received only 2% of global NCD research funding between 2012 and 2017 *(162)*. Multisectoral efforts addressing this imbalance is critical to ensuring that health resources are aligned with the level of disease burden.

5.2 Health monitoring systems are essential for standardizing CRD data collection and use

A robust and standardized health monitoring system is essential to accurately track the prevalence, burden, and epidemiological patterns of asthma and COPD, particularly in LMICs where data gaps are most pronounced. Despite asthma affecting over 260 million people and COPD ranking as the third leading cause of death globally *(161)*, many countries lack comprehensive surveillance systems to assess disease trends and inform health policies. Integrating CRD indicators into existing national and international health monitoring frameworks, such as the WHO STEPwise Approach to NCD Surveillance (WHO STEPS), can ensure systematic data collection of risk factors through patient questionnaires, physical measures, and biochemical measures, and create comparable data across regions *(163)*. Moreover, investing in capacity-building initiatives will enable LMICs to strengthen data infrastructure, ensuring reliable and disaggregated data on disease prevalence, risk factors, and healthcare access *(164)*. Without high-quality, real-time epidemiological data, governments and health systems risk underestimating the true burden of asthma and COPD, leading to insufficient resource allocation and missed opportunities for prevention and management. A coordinated global effort to enhance CRD surveillance will be pivotal in advancing evidence-based policies, reducing premature mortality, and improving health outcomes for millions affected worldwide.

5.3 Challenges in estimating the global burden of CRDs

Despite efforts to estimate the global prevalence and burden of CRDs, existing data remain highly variable and often unreliable due to methodological inconsistencies, limited population-based studies, and underdiagnosis in many regions. Current estimates, such as those from the Global Burden of Disease study, indicate that COPD affected over 213 million people and accounted for 3.72 million deaths worldwide in 2021, while asthma affected approximately 260 million people *(160)*. However, these figures rely on indirect modeling and extrapolation rather than actual numbers of cases, as many LMICs lack robust national surveillance systems to validate these projections with real-world epidemiological data *(164)*.

A major challenge in CRD epidemiology is the underdiagnosis and misclassification of COPD and asthma, particularly in primary health care settings where diagnostic tools like spirometry are not widely available *(165)*. Additionally, patient self-reported surveys often yield unreliable estimates, as symptoms of asthma and COPD overlap with other respiratory conditions, leading to overestimation or misclassification *(166)*. Differences in diagnostic criteria and definitions used across studies also contribute to inconsistencies in prevalence estimates, making cross-country comparisons difficult *(167)*.

Furthermore, many current CRD burden estimates fail to account for disparities in disease severity, healthcare access, and environmental exposures, particularly in regions with high levels of air pollution, biomass fuel exposure, and occupational hazards *(168)*. Without comprehensive, standardized surveillance systems, policymakers lack the granular data needed to develop targeted interventions, allocate resources effectively, and track the progress of CRD prevention and management strategies. Addressing these gaps requires greater investment in real-time epidemiological surveillance, standardized diagnostic criteria, and improved access to diagnostic tools, particularly in resource-limited settings where the burden of CRDs is expected to rise.

5.4 Establishing internationally recognized CRD indicators

The establishment of internationally recognized indicators at the health facility, sub-national, and national level is essential for accurately measuring the burden of CRDs and evaluating progress in improving care and reducing premature mortality. Unlike other major NCDs, such as cardiovascular diseases and cancer, asthma and COPD lack globally standardized reporting frameworks, leading to inconsistent tracking of disease prevalence, treatment outcomes, and healthcare access *(164, 169)*. The WHO STEPS, based on the collection of three steps of questionnaires, physical measures, and biochemical measures, provides a foundation for CRD monitoring, but policy changes to integrate specific asthma- and COPD-related indicators are needed to ensure that these diseases receive adequate policy attention *(163)*.

Robust CRD indicators should capture key metrics, including prevalence, incidence, hospitalization rates, medication availability, adherence to treatment guidelines, and mortality rates *(170)*. These indicators should be embedded within electronic health records and routine health information systems to enable real-time tracking and facilitate data-driven decision-making *(171)*. Additionally, disaggregating data by age, sex, socioeconomic status, and environmental exposures can help identify vulnerable populations and disparities in healthcare access *(166)*.

To maximize impact, governments and international health agencies must collaborate to integrate CRD indicators into existing NCD surveillance frameworks and national health information systems

(15). This will allow policymakers to benchmark progress, assess healthcare system performance, and ensure accountability in achieving global CRD targets. Furthermore, investing in capacity-building for LMICs will be essential to ensuring that these indicators are effectively implemented and utilized for policy development *(172)*. By prioritizing standardized, measurable indicators, decision-makers can drive evidence-based strategies that reduce premature deaths and improve quality of care for millions affected by asthma and COPD.

5.5 Continuous auditing mechanisms for asthma and COPD care

In addition to integrating CRD indicators into national and international health monitoring frameworks such as WHO STEPS, it is essential to implement continuous auditing mechanisms to assess the quality of asthma and COPD care in key healthcare settings. Regular auditing of asthma care in schools, utilizing the findings of the Global Asthma Network, can help evaluate and improve school-based asthma management programs. Similarly, ongoing assessments of asthma and COPD care in primary healthcare centers and outpatient hospital clinics, combined with onsite case-by-case teaching and mobile follow-up services, can strengthen patient outcomes and healthcare system performance. One critical step is auditing CRD-related curricula in nursing, pharmacy, and medical schools, particularly in LMICs, to ensure that training programs align with the latest evidence-based guidelines. By integrating up-to-date knowledge on asthma and COPD into medical education, countries can build a workforce that is better equipped to manage CRDs efficiently at all levels of care. These efforts must be accompanied by continuing professional development programs and hands-on training in healthcare facilities to reinforce best practices. These auditing strategies will provide policymakers with actionable insights to optimize CRD prevention and management at multiple levels.

5.6 Importance of translational and health system research for bridging gaps

Translational and health system research is crucial for bridging the gap, as emphasized in the WHO implementation roadmap for the Global Action Plan on NCDs *(173)*, which highlights implementation research as a key tool for adapting policies and interventions to diverse contexts between scientific discoveries and real-world implementation, ensuring that proven, cost-effective strategies for CRD prevention and management are effectively scaled, particularly in resource-constrained settings *(174)*. Despite the availability of evidence-based interventions, such as inhaled corticosteroids for asthma management, many LMICs struggle with poor access, weak health infrastructure, and policy implementation gaps *(175)*. Investing in translational research can help adapt global clinical guidelines to local contexts, addressing health system barriers that hinder the widespread adoption of best-practice treatments and interventions *(176)*.

5.7 Strengthening policy formulation through health system research

Health system research further strengthens policy formulation and implementation by identifying bottlenecks in service delivery, healthcare financing, and workforce capacity *(177)*. Recent priority-

setting exercises, such as the global e-Delphi process of the International Primary Care Respiratory Group, involving detailed discussions between experts, and the expert consultation led by Adeloye et al., have identified key research directions to address the CRD burden *(178, 179)*. These include improving the diagnosis and self-management of asthma and COPD, tobacco cessation strategies, chronic cough, and multimorbidity in primary care. Other directions involve expanding access to pulmonary rehabilitation and developing effective screening tools for COPD in low-resource settings and implementation by identifying bottlenecks in service delivery, healthcare financing, and workforce capacity *(177)*. Studies show that integrating CRD management into primary health care reduces hospital admissions and improves patient outcomes, yet systemic inefficiencies often delay the adoption of such models *(164)*. Additionally, cost-effectiveness studies play a pivotal role in gaining government investments by demonstrating how early detection and affordable treatment options can reduce long-term healthcare expenditures and economic burden *(180)*.

5.8 Funding and expanding health system research for scalable solutions

A concerted effort to fund and expand health system research will allow policymakers to identify scalable and sustainable solutions, ensuring that CRD management strategies are contextually appropriate, financially viable, and effectively integrated into national health programs *(181)*. Strengthening research collaborations between academic institutions, governments, and global health agencies can further drive evidence-based policymaking, ultimately improving CRD prevention, diagnosis, and treatment worldwide.

A strong example of research-informed policy driving expanded coverage for COPD management at the primary care level is the National Basic Public Health Service (NBPHS) program of China. Launched in 2009, the NBPHS represents a long-standing government commitment to improving equity in access to essential public health services, including for CRDs such as COPD *(182)*. The program has integrated COPD screening and management into primary care through standard protocols, health education, and follow-up services, particularly targeting older adults and high-risk populations *(182)*. This approach was informed by epidemiological studies showing a rising burden of COPD in China and the need for early diagnosis and preventive interventions available in easy-to-access community settings *(183)*. Evidence suggests that embedding COPD management into routine public health services under NBPHS has led to improved COPD case detection and treatment coverage, and offers a model for other countries to replicate to close equity gaps in CRD care *(182)*. The integration of research findings into national policy frameworks such as NBPHS of China highlights the importance of robust data systems and state-led strategies in shaping health service delivery for CRDs.

5.9 Addressing CRD disparities through multisectoral action and inclusive policy

A multisectoral approach is essential to address disparities in CRD outcomes, particularly in vulnerable and crisis-affected populations. Integrating CRD prevention and care into humanitarian response frameworks, refugee health programs, and global climate action policies ensures that these efforts reflect the realities of at-risk populations *(184, 185)*. However, a major barrier remains the lack of high-quality, disaggregated, and longitudinal data on CRDs, especially in LMICs, where underdiagnosis, weak surveillance systems, and fragmented data collection persist *(186)*. Investing in

research that evaluates the impact of environmental and social determinants (such as air pollution, displacement, and housing quality) on CRD burden is critical to designing targeted, evidence-based interventions *(187)*. Reference models from LMICs demonstrate how integrating environmental data with health outcomes can enhance forecasting and policy alignment, especially when linked to early-warning systems and climate resilience planning *(184, 187)*. Furthermore, implementation research is needed to adapt successful CRD strategies to diverse contexts, ensuring sustainability and equity. To address this, stakeholders should prioritize not only the implementation of integrated care models but also the real-time documentation and evaluation of such interventions, especially during pilot phases in fragile settings, to generate actionable evidence and bridge the current knowledge gap *(185)*. Prioritizing CRDs within broader political and humanitarian agendas aligns with the goals of universal health coverage and strengthens global health equity efforts *(188)*.

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